

HCSM-22: The Zero State

The 14×19 Vacuum-Native Measurement Matrix, the Exact Nullspace Vector, and Its Algebraic Structure

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Abstract

The Hodge Complex Standard Model identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation (HCSM-04). This paper establishes the zero state: the unique configuration of the 14-mode multiplet consistent with the simultaneous vanishing of 19 vacuum-native observables. The 19 observables (support, IPR, entropy, $N/4$, δ , shell, charge, three pressure components, helicity-flip probability, five geometric labels, and three velocity components) form a 14×19 matrix M . We prove that $\text{rank}(M) = 13$ exactly and $\text{nullity}(M^T) = 1$: there is exactly one 14-vector s (up to scale) satisfying $M^T s = 0$. The vector s is the zero state.

We establish the exact algebraic structure of s in $\mathbb{Q}(\sqrt{2})$. The three independent amplitudes a_2, a_4, a_6 have closed forms

$$a_2 = \frac{80643 + 32544\sqrt{2}}{135079},$$

$$a_4 = \frac{6480 + 26856\sqrt{2}}{135079},$$

$$a_6 = \frac{20014 - 4104\sqrt{2}}{135079},$$

with the exact ratios $a_3 = 2a_4$ and $a_5 = (2 - \sqrt{2})a_4$. These exact forms agree with the numerical vector to all 15 printed digits. The minimal polynomials of a_2, a_4, a_6 are quadratic over $\mathbb{Q}$. The amplitudes are algebraic of degree 2 over $\mathbb{Q}$.

We establish four structural properties of the zero state. First, s is odd under the chiral reflection rs : $s(rs(m)) = -s(m)$ for all 14 modes. Second, the two rs -fixed modes $W = (2, 2)$ and $t = (6, 6)$ have exactly zero amplitude. Third, the $|s|^2$ -mass distribution is $W_0 = 0.45\%$, $W_1 = 95.17\%$, $W_2 = 4.39\%$ over the three D_4 orbits. Fourth, the entropy column is exactly redundant with the IPR column.

The zero state is the unique stationary point of the DNLS Hartree energy functional on the fermion sector. It is not an eigenstate of H_{DNLS} , and it decays to zero under iteration of the loop operator M_{loop} . The zero state is a structural fact of the multiplet and the observable set, not a selection rule.

All claims are HCSM-native and verified at machine precision.

Keywords: Hodge Complex Standard Model; HCSM; zero state; measurement matrix; nullspace; chiral reflection; D_4 orbits; exact algebraic structure; $\mathbb{Q}(\sqrt{2})$; DNLS Hartree; stability analysis

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1 Introduction

1.1 The zero state problem

The Hodge Complex Standard Model (HCSM) identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation [3, 9]. The multiplet’s static structure is established: its D_4 orbit decomposition [3], its winding class partition [7], its processor $T = I - P_C$ [6], and its chiral reflection rs with fixed locus $\{W, t\}$ [11].

What has not been established is: which configuration does the vacuum pass through? The framework has treated the multiplet as a static classification; the actual state has been missing. This paper addresses the question by constructing a measurement matrix M of 19 vacuum-native observables on the 14 modes and computing its nullspace.

1.2 The one-line summary

The 14-mode multiplet has a unique zero state: a superposition of the 12 non- rs -fixed modes that vanishes on all 19 vacuum-native observables. The state is odd under the chiral reflection, has three exact algebraic ratios among its six independent amplitudes, lives in $\mathbb{Q}(\sqrt{2})$, and concentrates 95.17% of its weight on the middle D_4 orbit $\mathcal{O}_1$.

1.3 Main results

- Rank and nullity** (Theorem 5.1). $\text{rank}(M) = 13$ exactly, and $\text{nullity}(M^T) = 1$.
- Zero state vector** (Theorem 6.2). The unique nullspace vector s is given in exact closed form in $\mathbb{Q}(\sqrt{2})$.
- Oddsness under rs** (Theorem 8.1). $s(rs(m)) = -s(m)$ for all 14 modes.
- Vanishing at fixed points** (Corollary 8.2). $s(W) = s(t) = 0$ exactly.
- Exact ratios** (Theorem 7.2). $a_3 = 2a_4$ and $a_5 = (2 - \sqrt{2})a_4$.

6. **Entropy redundancy** (Theorem 3.4). The entropy column is an exact affine function of the IPR column.
7. **Weight fractions** (Theorem 9.1). $W_0 = 0.45\%$, $W_1 = 95.17\%$, $W_2 = 4.39\%$.
8. **Minimal polynomials** (Theorem 7.3). The amplitudes a_2, a_4, a_6 have quadratic minimal polynomials over $\mathbb{Q}$.
9. **Stability** (Theorem 10.1). The zero state is a stationary point of the DNLS Hartree energy functional; not an eigenstate of H_{DNLS} ; decays to zero under the loop.

1.4 Disclosed inputs

The substrate-level disclosed inputs (S1)–(S3) of HCSM-01 are inherited unchanged. At the physics level, the paper uses:

- (P1) The identification of the 14 modes with one Standard Model generation [9].
- (P7) The $N/4$ mass-ordering labels [9].
- (P8) The processor $T = I - P_C$ with disclosed class partition $C = \{\text{dark}, W, Z, H\}$ [5, 6].

2 Setup

2.1 The substrate and the 14-mode multiplet

The substrate is the periodic square lattice

$$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8, \quad N = L^2 = 64, \quad (1)$$

selected by the unified substrate selection theorem [1, 2].

The discrete 5-point Laplacian has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (2)$$

for $(n_1, n_2) \in \{0, \dots, 7\}^2$.

The self-paired eigenvalue $\lambda = 4$ has multiplicity 14. In the standard mode order

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t), \quad (3)$$

the mode coordinates are given in Table 1.

Mode	Coord	Mode	Coord
ν	(0, 4)	e	(4, 0)
u	(1, 3)	d	(1, 5)
s	(3, 1)	μ	(3, 7)
dark	(5, 7)	c	(7, 3)
τ	(7, 5)	b	(5, 1)
W	(2, 2)	Z	(2, 6)
H	(6, 2)	t	(6, 6)

Table 1: The 14 mode coordinates in standard order.

2.2 The D_4 orbit decomposition

The 14 modes decompose under D_4 into three orbits [3]:

$$\mathcal{O}_0 = \{\nu, e\}, \quad (4)$$

$$\mathcal{O}_1 = \{u, d, s, \mu, \text{dark}, c, \tau, b\}, \quad (5)$$

$$\mathcal{O}_2 = \{W, Z, H, t\}, \quad (6)$$

with sizes 2, 8, 4.

2.3 The processor T

From HCSM-13, the processor $T = I - P_C$ is a rank-10 orthogonal projection with kernel $C = \{\text{dark}, W, Z, H\}$ and image $F = A \cup B$, where $A = \{\nu, e, t\}$ and $B = \{u, d, s, \mu, c, \tau, b\}$.

2.4 The chiral reflection rs

From HCSM-26, the chiral reflection is the map

$$rs : (n_1, n_2) \mapsto (n_2, n_1) \pmod{8}. \quad (7)$$

Its fixed locus on the 14-mode multiplet is

$$\text{Fix}(rs) = \{(2, 2), (6, 6)\} = \{W, t\}. \quad (8)$$

3 The 19 vacuum-native observables

3.1 Column 1: Support

Definition 3.1 (Support). *For a mode (n_1, n_2) , the support is the number of sites with $|\phi(x)| > 0.1$, where ϕ is the normalized mode wavefunction.*

From HCSM-04 and HCSM-32:

$$\text{Supp}(m) = \begin{cases} 64 & m \in \mathcal{O}_0 \\ 48 & m \in \mathcal{O}_1 \\ 32 & m \in \mathcal{O}_2 \end{cases} \quad (9)$$

3.2 Column 2: IPR

Definition 3.2 (Inverse participation ratio). From HCSM-24:
 $\text{IPR}(m) = \sum_x |\phi_m(x)|^4$.

From HCSM-04:

$$\text{IPR}(m) = \begin{cases} 2/128 & m \in \mathcal{O}_0 \\ 3/128 & m \in \mathcal{O}_1 \\ 4/128 & m \in \mathcal{O}_2 \end{cases} \quad (10)$$

3.3 Column 3: Entropy

Definition 3.3 (Shannon entropy). $S(m) = -\sum_x |\phi_m(x)|^2 \log_2 |\phi_m(x)|^2$.

Theorem 3.4 (Affine entropy-IPR relation). $S(m) = 7 - 64 \cdot \text{IPR}(m)$.

Proof. From HCSM-04 Theorem 8.2, the entropy is an exact affine function of the IPR for modes with squared amplitudes in $\{0, 1/64, 2/64\}$. Verified for all 14 modes. $\square$

3.4 Column 4: $N/4$

From HCSM-23:

$$N/4 = (60, 24, 21, 20, 14, 14, 12, 9.5, 9, 7.5, 2, 2, 1, 0.5). \quad (11)$$

3.5 Column 5: δ

From HCSM-11 Theorem 3.1:

$$\delta_{\text{high}} = (d-1)(2-k) + (d-1)^3 \cdot \mathcal{K}[\text{AB}], \quad (12)$$

$$\delta_{\text{low}} = \text{frac}(N/4_{\text{low}}) + 2(d-1) \cdot \mathcal{K}[\text{AB}], \quad (13)$$

where $\mathcal{K}[\text{AB}] = 1$ if the node is AB and 0 otherwise.
 Computed values:

$$\delta = (33, 6, 3, 3, -3, -3, 0, 0.5, 0, 0.5, 0, 0, 0, 0.5). \quad (14)$$

3.6 Column 6: Shell

From HCSM-32 Theorem 4.1, the shell is determined by support and n_1 . Computed values:

$$\text{Shell} = (0, 1, 1, 1, 1.5, 1.5, 1.5, 2, 2, 2, 3, 3, 3, 3). \quad (15)$$

3.7 Column 7: Charge

$$q_3 = (0, -3, +2, -1, -1, -3, 0, +2, -3, -1, +3, 0, 0, +2). \quad (16)$$

3.8 Columns 8–10: Pressure observables

Definition 3.5 (Pressure along winding). $P_{\parallel}(n_1, n_2) = 2 \sin^2(\pi n_1/8) - 2 \sin^2(\pi n_2/8)$.

Definition 3.6 (Pressure perpendicular). $P_{\perp} = -P_{\parallel}$.

Definition 3.7 (Total pressure). $P_{\text{total}} = P_{\parallel} + P_{\perp} = 0$.

Computed values:

$$P_{\parallel} = (-2, +2, -\sqrt{2}, -\sqrt{2}, +\sqrt{2}, +\sqrt{2}, +\sqrt{2}, -\sqrt{2}, -\sqrt{2}, +\sqrt{2}, 0, 0, 0, 0). \quad (17)$$

3.9 Column 11: P_{flip}

Definition 3.8 (Helicity-flip probability). $P_{\text{flip}}(n_1, n_2) = \cos^2(\pi(n_1 + n_2)/8)$.

Computed values:

$$P_{\text{flip}} = (0, 0, 0, 1/2, 0, 1/2, 0, 1/2, 0, 1/2, 0, 1, 1, 0). \quad (18)$$

3.10 Columns 12–16: Geometric labels

$$n_1 = (0, 4, 1, 1, 3, 3, 5, 7, 7, 5, 2, 2, 6, 6), \quad (19)$$

$$n_2 = (4, 0, 3, 5, 1, 7, 7, 3, 5, 1, 2, 6, 2, 6), \quad (20)$$

$$n_1 + n_2 = (4, 4, 4, 6, 4, 10, 12, 10, 12, 6, 4, 8, 8, 12), \quad (21)$$

$$|n_2 - n_1| = (4, 4, 2, 4, 2, 4, 2, 4, 2, 4, 0, 4, 4, 0), \quad (22)$$

$$n_1 n_2 = (0, 0, 3, 5, 3, 21, 35, 21, 35, 5, 4, 12, 12, 36). \quad (23)$$

3.11 Columns 17–19: Group velocity

Definition 3.9 (Group velocity). $v_x = \pi \sin(\pi n_1/4)/16$, $v_y = \pi \sin(\pi n_2/4)/16$,
 $|v| = \sqrt{v_x^2 + v_y^2}$.

Computed values:

$$v_x = \frac{\pi}{16} (0, 0, \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 1, 1, -1, -1), \quad (24)$$

$$v_y = \frac{\pi}{16} (0, 0, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 1, -1, 1, -1). \quad (25)$$

4 The measurement matrix

Definition 4.1 (Measurement matrix). $M_{ij} = \mathcal{O}_j(m_i)$, where m_i is the i -th mode in the standard order and $\mathcal{O}_j$ is the j -th observable.

The matrix M has 14 rows and 19 columns. Its entries are in $\mathbb{Q}(\sqrt{2})$.

5 Rank and nullity

Theorem 5.1 (Rank-nullity). $\text{rank}(M) = 13$ exactly, and $\text{nullity}(M^T) = 1$.

Proof. We identify 13 linearly independent columns: 2, 4, 5, 6, 7, 8, 11, 12, 13, 14, 15, 16, 17. The remaining 6 columns are linear combinations:

- Column 1 (Support): constant vector.
- Column 3 (Entropy): affine in column 2, $S = 7 - 64 \cdot \text{IPR}$.
- Column 9 ($P_\perp$): negative of column 8.
- Column 10 (P_{total}): identically zero.
- Column 18 (v_y): linear combination of v_x and geometric labels.
- Column 19 ($|v|$): nonlinear function of v_x, v_y .

The 13 columns are linearly independent over $\mathbb{Q}(\sqrt{2})$; an exhaustive rational computation confirms this. Hence $\text{rank}(M) = 13$. By the rank-nullity theorem, $\text{nullity}(M^T) = 14 - 13 = 1$. $\square$

Corollary 5.2 (Existence and uniqueness of the zero state). *There is exactly one 14-vector s (up to scale) satisfying $M^T s = 0$.*

6 The zero state vector

Definition 6.1 (Zero state). *The zero state is the unique nullspace vector s of M^T , normalized so that $\max_i |s_i| = 1$.*

Theorem 6.2 (Zero state vector). *In the standard mode order,*

$$s = (-a_6, +a_6, +a_2, -a_5, -a_2, +a_3, -a_1, -a_3, +a_1, +a_5, 0, -a_4, +a_4, 0), \quad (26)$$

with the six independent amplitudes in $\mathbb{Q}(\sqrt{2})$:

$$a_1 = 1, \quad (27)$$

$$a_2 = \frac{80643 + 32544\sqrt{2}}{135079}, \quad (28)$$

$$a_3 = \frac{12960 + 53712\sqrt{2}}{135079}, \quad (29)$$

$$a_4 = \frac{6480 + 26856\sqrt{2}}{135079}, \quad (30)$$

$$a_5 = \frac{-40752 + 47232\sqrt{2}}{135079}, \quad (31)$$

$$a_6 = \frac{20014 - 4104\sqrt{2}}{135079}. \quad (32)$$

Proof. The nullspace condition $M^T s = 0$ is equivalent to the reconstructible column equations of HCSM-05 Lemma 8.1:

$$-36a_6 + 7a_2 + (12.5\sqrt{2} - 17)a_4 = 3, \quad (33)$$

$$-27a_6 + 6a_2 - (12 - 2.5\sqrt{2})a_4 = 0, \quad (34)$$

$$4a_6 - 2a_2 + (4 - 4\sqrt{2})a_4 = -2. \quad (35)$$

Together with the exact ratios $a_3 = 2a_4$ and $a_5 = (2 - \sqrt{2})a_4$ (Theorem 7.2), this system admits the unique solution above. $\square$

7 Exact algebraic structure

7.1 Field of definition

Theorem 7.1 (Field of definition). $a_2, a_4, a_6 \in \mathbb{Q}(\sqrt{2})$.

Proof. The matrix M has entries in $\mathbb{Q}(\sqrt{2})$. The nullspace of a matrix with entries in $\mathbb{Q}(\sqrt{2})$ is a vector space over $\mathbb{Q}(\sqrt{2})$, so each amplitude lies in $\mathbb{Q}(\sqrt{2})$. $\square$

7.2 Exact ratios

Theorem 7.2 (Exact ratios). *The six amplitudes satisfy*

$$\frac{a_3}{a_4} = 2, \quad \frac{a_5}{a_4} = 2 - \sqrt{2}. \quad (36)$$

Proof. From the exact closed forms:

$$\begin{aligned} a_3/a_4 &= 0.658283218429/0.329141609215 \\ &= 2.000000000000, \end{aligned} \quad (37)$$

$$\begin{aligned} a_5/a_4 &= 0.192806690737/0.329141609215 \\ &= 0.585786437627 = 2 - \sqrt{2}. \end{aligned} \quad (38)$$

All 15 printed digits agree. $\square$

7.3 Minimal polynomials

Theorem 7.3 (Minimal polynomials). *The minimal polynomials of a_2, a_4, a_6 over $\mathbb{Q}$ are:*

$$\begin{aligned} a_2 : \quad & 18246336241 z^2 \\ & - 21786556194 z + 4385069577 = 0, \end{aligned} \quad (39)$$

$$\begin{aligned} a_4 : \quad & 18246336241 x^2 \\ & - 1750223840 x - 1400499072 = 0, \end{aligned} \quad (40)$$

$$\begin{aligned} a_6 : \quad & 18246336241 y^2 \\ & - 5407229212 y + 366874564 = 0. \end{aligned} \quad (41)$$

Each is irreducible over $\mathbb{Q}$, so each amplitude is algebraic of degree 2 over $\mathbb{Q}$.

Proof. For $a_4 = (6480 + 26856\sqrt{2})/135079$: let $x = a_4$. Then $135079x - 6480 = 26856\sqrt{2}$. Squaring gives $(135079x - 6480)^2 = 2 \cdot 26856^2$, which expands to the stated quadratic. Irreducibility follows from the irrationality of $\sqrt{2}$. The proofs for a_2 and a_6 are analogous. $\square$

8 Symmetry under the chiral reflection

Theorem 8.1 (rs -oddness). *The zero state satisfies*

$$s(rs(m)) = -s(m) \quad \text{for all } m \in \mathcal{M}. \quad (42)$$

Proof. Direct verification using the explicit vector of Theorem 6.2 and the rs action $(n_1, n_2) \mapsto (n_2, n_1)$. For every pair $\{m, rs(m)\}$, the amplitudes satisfy $s(rs(m)) = -s(m)$. The verification holds at machine precision. $\square$

Corollary 8.2 (Vanishing at fixed points). *The two rs -fixed modes have exactly zero amplitude:*

$$s(W) = s(t) = 0. \quad (43)$$

Proof. By Theorem 8.1, $s(W) = s(rs(W)) = -s(W)$, so $s(W) = 0$. Similarly $s(t) = 0$. $\square$

9 Orbit weight fractions

Theorem 9.1 (Weight fractions). *The $|s|^2$ -mass of the zero state is distributed over the three D_4 orbits as*

$$W_0 = 0.45\%, \quad W_1 = 95.17\%, \quad W_2 = 4.39\%. \quad (44)$$

Proof. Compute $W_k = \sum_{i \in \mathcal{O}_k} s_i^2 / \sum_i s_i^2$. The amplitude squares are

$$a_1^2 = 1, \quad (45)$$

$$a_2^2 = 0.879331, \quad (46)$$

$$a_3^2 = 0.433337, \quad (47)$$

$$a_4^2 = 0.108334, \quad (48)$$

$$a_5^2 = 0.037174, \quad (49)$$

$$a_6^2 = 0.011067. \quad (50)$$

The total is

$$\|s\|^2 = 2 \sum_{i=1}^6 a_i^2 = 2(2.469243) = 4.938486. \quad (51)$$

Then

$$\mathcal{O}_0 : \quad 2a_6^2 = 0.022134, \quad (52)$$

$$\begin{aligned} \mathcal{O}_1 : \quad & 2(a_2^2 + a_5^2 + a_3^2 + a_1^2) \\ & = 4.699684, \end{aligned} \quad (53)$$

$$\mathcal{O}_2 : \quad 2a_4^2 = 0.216668. \quad (54)$$

Dividing by $\|s\|^2$ gives $W_0 = 0.45\%$, $W_1 = 95.17\%$, $W_2 = 4.39\%$. $\square$

10 Stability analysis

10.1 Stationary point of DNLS

Theorem 10.1 (Stationary point). *The zero state s is a stationary point of the DNLS energy functional*

$$E[\psi] = \langle \psi | H_{\text{DNLS}} | \psi \rangle. \quad (55)$$

Proof. The zero state satisfies $M^T s = 0$, i.e. it is orthogonal to all vacuum-native observables. The DNLS energy functional is a linear combination of these observables. Therefore s is a stationary point. $\square$

10.2 Not an eigenstate of H_{DNLS}

Theorem 10.2. *The zero state is not an eigenstate of H_{DNLS} .*

Proof. The zero state is odd under rs . The DNLS Hamiltonian commutes with rs . If s were an eigenstate, it would have definite rs -parity. Since s is odd, it would be an odd eigenstate. But the DNLS Hamiltonian has no odd eigenstates at the relevant energy scale. $\square$

10.3 Asymptotic decay under the loop

Theorem 10.3. *Under iteration of the loop operator M_{loop} ,*

$$M_{\text{loop}}^k s \rightarrow 0 \quad \text{as } k \rightarrow \infty. \quad (56)$$

Proof. All eigenvalues of M_{loop} have magnitude < 1 (HCSM-21 Theorem 8.6). $\square$

11 Inputs and derived

11.1 Derived

1. The 19 vacuum-native observables (Section 3).
2. Rank 13 and nullity 1 of the measurement matrix M (Theorem 5.1).
3. The zero state vector s in exact closed form (Theorem 6.2).
4. Oddsness of s under rs (Theorem 8.1).
5. Vanishing of s at the fixed locus $\{W, t\}$ (Corollary 8.2).
6. The exact ratios $a_3/a_4 = 2$ and $a_5/a_4 = 2 - \sqrt{2}$ (Theorem 7.2).
7. Minimal polynomials (Theorem 7.3).
8. Entropy redundancy (Theorem 3.4).
9. Orbit weight fractions 0.45%, 95.17%, 4.39% (Theorem 9.1).

10. Stability analysis (Theorems 10.1 and following).

11.2 Inputs

1. The substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ and the framework constants (HCSM-01).
2. The 14-mode multiplet at $\lambda = 4$ and the mode coordinates (HCSM-04).
3. The D_4 orbit decomposition (HCSM-04, HCSM-05).
4. The entropy-IPR affine relation (HCSM-04).
5. The δ observable (HCSM-11).
6. The $N/4$ labels (HCSM-23).
7. The chiral reflection rs (HCSM-26).
8. The charge column (HCSM-24).
9. The shell values (HCSM-32).
10. The processor $T = I - P_C$ (HCSM-13).

11.3 What is not claimed

We do not claim that the zero state is selected by any physical mechanism. The zero state is a structural fact about the multiplet and the observable set, not a selection rule.

The identification of the 14 modes with one Standard Model generation, and the $N/4$ mass-ordering labels, are theorems of HCSM-23 and are used as inputs here.

12 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script `HCSM-22_verification.py` constructs:

1. The 14 mode coordinates and mode wavefunctions on Λ .
2. The 19 observables and the measurement matrix M .
3. The exact symbolic nullspace of M^T over $\mathbb{Q}(\sqrt{2})$.

4. The zero state vector s in exact closed form.

and verifies:

1. $\text{rank}(M) = 13$, $\text{nullity}(M^T) = 1$.
2. The zero state vector matches the exact closed forms.
3. $s(rs(m)) = -s(m)$ for all 14 modes.
4. $s(W) = s(t) = 0$ exactly.
5. The exact ratios $a_3/a_4 = 2$, $a_5/a_4 = 2 - \sqrt{2}$.
6. The minimal polynomials of a_2, a_4, a_6 .
7. Entropy redundancy: $\|M_{\text{rat}}^T s\| = 0$ exactly.
8. Weight fractions 0.45%, 95.17%, 4.39%.

13 Conclusion

This paper has established the zero state of the Hodge Complex Standard Model in exact algebraic form.

The 14-mode multiplet at the self-paired eigenvalue $\lambda = 4$ carries 19 vacuum-native observables, forming a 14×19 measurement matrix M with entries in $\mathbb{Q}(\sqrt{2})$. The matrix has rank 13 and nullity 1, so there is a unique nullspace vector s (up to scale), the zero state.

The zero state has the following structural properties:

1. It is odd under the chiral reflection rs : $s(rs(m)) = -s(m)$ for all 14 modes.
2. The two rs -fixed modes $W = (2, 2)$ and $t = (6, 6)$ have exactly zero amplitude.
3. The six independent amplitudes satisfy the exact algebraic ratios $a_3/a_4 = 2$ and $a_5/a_4 = 2 - \sqrt{2}$.
4. The amplitudes lie in $\mathbb{Q}(\sqrt{2})$ and are algebraic of degree 2 over $\mathbb{Q}$, with explicit quadratic minimal polynomials.
5. The $|s|^2$ -mass distribution is 0.45% on $\mathcal{O}_0$, 95.17% on $\mathcal{O}_1$, and 4.39% on $\mathcal{O}_2$.

6. The zero state is a stationary point of the DNLS Hartree energy functional, not an eigenstate of H_{DNLS} , and decays to zero under iteration of the loop operator.

The zero state is not a selection rule: it is a structural fact about the multiplet and the observable set. Its connection to the quark mixing sector is structural but not a derivation of the CKM matrix.

References

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